

Student: Jonatan Rabjerg
Date: 1/15/24

Instructor: Andreas Basse-O'Connor
Course: Calculus Beta

Assignment: MyLab Math - Uge 9

The digestion time (in hours) of a fixed amount of food is exponentially distributed with $a = 1$.

- a. Find the mean digestion time.
- b. Find the probability that the digestion time is less than 192 minutes.

a. Recall that if X is a random variable with probability density function $f(x) = ae^{-ax}$ for x in $[0, \infty)$, then $\mu = \frac{1}{a}$.

The given value of a is 1.

Substitute the given value of a into the equation and simplify to find μ .

$$\mu = \frac{1}{1} = 1$$

The mean digestion time is 1 hour.

b. The probability distribution function is $f(t) = e^{-t}$.

To find the probability, first convert 192 minutes to hours.

$$192 \text{ minutes} = 3.2 \text{ hours}$$

Now set up the integral.

$$P(T \leq 3.2 \text{ hours}) = \int_0^{3.2} e^{-t} dt$$

Take the antiderivative.

$$\int_0^{3.2} e^{-t} dt = -e^{-t} \Big|_0^{3.2}$$

Evaluate the antiderivative.

$$-e^{-t} \Big|_0^{3.2} = -e^{-3.2} - (-e^0) \approx 0.9592$$

Therefore, the probability that the digestion time is less than 192 minutes is 0.9592.